

Yifan Chen

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Updated May, 2026

EMPLOYMENT

Department of Mathematics, UCLA

Assistant Professor (tenure track)

Los Angeles, CA

2025/7–present

DataX, UCLA

Affiliated Faculty

Los Angeles, CA

2025/7–present

Courant Institute, New York University

Assistant Professor/Courant Instructor (non-tenure track)

New York, NY

2023/9–2025/7

EDUCATION

California Institute of Technology

Ph.D. in Applied and Computational Mathematics

Advisors: Thomas Y. Hou, Houman Owhadi, Andrew M. Stuart

Pasadena, California

2018–2023

Tsinghua University

B.S. in Pure and Applied Mathematics

Beijing, China

2014–2018

RESEARCH INTERESTS

My research lies at the intersection of applied and computational mathematics, applied probability, and statistics. I am interested in the mathematical analysis and applications of probabilistic inference, stochastic algorithms, and PDEs in scientific computing and data science. My recent focus is on sampling and generative modeling.

PUBLICATIONS

Preprints

- [1] **Yifan Chen**, Eric Vanden-Eijnden. Variational Optimality of Föllmer Processes in Generative Diffusions. *arxiv: 2602.10989*, 2026.
- [2] Baojun Che, **Yifan Chen**, Daniel Zhengyu Huang, Xinying Mao, Weijie Wang Adaptive Exponential Integration for Stable Gaussian Mixture Black-Box Variational Inference. *arXiv:2601.14855*, 2026.
- [3] **Yifan Chen**, Sifan Liu. Rotated Mean-Field Variational Inference and Iterative Gaussianization. *arxiv: 2510.07732*, 2025
- [4] **Yifan Chen**, Eric Vanden-Eijnden. Scale-Adaptive Generative Flows for Multiscale Scientific Data. *arxiv: 2509.02971*, 2025
- [5] **Yifan Chen**, Eric Vanden-Eijnden, Jiawei Xu. Lipschitz-Guided Design of Interpolation Schedules in Generative Models. *arxiv: 2509.01629*, 2025
- [6] **Yifan Chen**. New Affine Invariant Ensemble Samplers and Their Dimensional Scaling. *arxiv: 2505.02987*, 2025

- [1] José A. Carrillo, **Yifan Chen**, Daniel Zhengyu Huang, Jiaoyang Huang, Dongyi Wei. Fisher-Rao Gradient Flow: Geodesic Convexity and Functional Inequalities. *SIAM Journal on Mathematical Analysis*, 2026
- [2] **Yifan Chen**, Xiaouu Cheng, Jonathon Niles-Weed, Jonathon Weare. Convergence of Unadjusted Langevin in High Dimensions: Delocalization of Bias. *Communications on Pure and Applied Mathematics*, 2026.
- [3] **Yifan Chen**, Daniel Zhengyu Huang, Jiaoyang Huang, Sebastian Reich, Andrew M. Stuart. Sampling via Gradient Flows in the Space of Probability Measures. *Mathematics of Computation*, 2026
- [4] Huan Zhang, **Yifan Chen**, Eric Vanden-Eijnden, Benjamin Peherstorfer. Sequential-in-time training of nonlinear parametrizations for solving time-dependent partial differential equations. *SIAM Review Research Spotlights*, 2026.
- [5] Baojun Che, **Yifan Chen**, Zhenghao Huan, Daniel Zhengyu Huang, Weijie Wang. Stable Derivative Free Gaussian Mixture Variational Inference for Bayesian Inverse Problems. *SIAM Journal on Scientific Computing*, 2025.
- [6] **Yifan Chen**, Ethan N. Epperly, Joel A. Tropp, and Robert J. Webber. Randomly pivoted Cholesky: Practical approximation of a kernel matrix with few entry evaluations. *Communications on Pure and Applied Mathematics*, 2024.
- [7] **Yifan Chen**, Bamdad Hosseini, Houman Owhadi, Andrew M. Stuart. Gaussian Measures Conditioned on Nonlinear Observations: Consistency, MAP Estimators, and Simulation. *Statistics and Computing*, 2024.
- [8] **Yifan Chen**, Daniel Zhengyu Huang, Jiaoyang Huang, Sebastian Reich, Andrew M. Stuart. Efficient, Multimodal, and Derivative-Free Bayesian Inference With Fisher-Rao Gradient Flows. *Inverse Problems*, 2024
- [9] Pau Batlle, **Yifan Chen**, Bamdad Hosseini, Houman Owhadi, Andrew M. Stuart. Error Analysis of Kernel/GP Methods for Nonlinear and Parametric PDEs. *Journal of Computational Physics*, 2024.
- [10] Yu Sun, Zihui Wu, **Yifan Chen**, Berthy T. Feng, Katherine L. Bouman. Provable Probabilistic Imaging using Score-Based Generative Priors. *IEEE Transactions on Computational Imaging*, 2024.
- [11] **Yifan Chen**, Houman Owhadi, Florian Schaefer. Sparse Cholesky Factorization for Solving Nonlinear PDEs via Gaussian Processes. *Mathematics of Computation*, 2024.
- [12] **Yifan Chen**, Thomas Y. Hou, and Yixuan Wang. Exponentially convergent multiscale methods for 2D high frequency heterogeneous Helmholtz equations. *SIAM Multiscale Modeling & Simulation*, 21(3): 849–883, 2023.
- [13] **Yifan Chen**, Thomas Y. Hou, and Yixuan Wang. Exponentially convergent multiscale finite element method. *Communications on Applied Mathematics and Computation*, 1–17, 2023.
- [14] **Yifan Chen** and Thomas Y. Hou. Multiscale elliptic PDE upscaling and function approximation via subsampled data. *SIAM Multiscale Modeling & Simulation*, 20(1):188–219, 2022.
- [15] **Yifan Chen**, Bamdad Hosseini, Houman Owhadi, and Andrew M. Stuart. Solving and learning nonlinear PDEs with Gaussian processes. *Journal of Computational Physics*, 447:110668, 2021.
- [16] **Yifan Chen**, Houman Owhadi, and Andrew M. Stuart. Consistency of empirical Bayes and kernel flow for hierarchical parameter estimation. *Mathematics of Computation*, 90(332):2527–2578, 2021.
- [17] **Yifan Chen**, Thomas Y. Hou, and Yixuan Wang. Exponential convergence for multiscale linear elliptic PDEs via adaptive edge basis functions. *SIAM Multiscale Modeling & Simulation*, 19(2):980–1010, 2021.
- [18] **Yifan Chen** and Thomas Y. Hou. Function approximation via the subsampled Poincaré inequality. *Discrete & Continuous Dynamical Systems-A*, 41(1), 2021.
- [19] **Yifan Chen** and Wuchen Li. Optimal transport natural gradient for statistical manifolds with continuous sample space. *Information Geometry*, 3(1):1–32, 2020.
- [20] **Yifan Chen**, Yuejiao Sun, and Wotao Yin. Run-and-Inspect Method for nonconvex optimization and global optimality bounds for R-local minimizers. *Mathematical Programming*, 176(1): 39–67, 2019.

- [21] Jing Chen, **Yifan Chen**, Hao Wu, and Dinghui Yang. The quadratic Wasserstein metric for earthquake location. *Journal of Computational Physics*, 373:188–209, 2018.

Conference publications

- [1] Wenda Chu, Zihui Wu, **Yifan Chen**, Yang Song, Yisong Yue. Split Gibbs Discrete Diffusion Posterior Sampling. *NeurIPS*, 2025
- [2] **Yifan Chen**, Mark Goldstein, Mengjian Hua, Michael S. Albergo, Nicholas M. Boffi, Eric Vanden-Eijnden. Probabilistic Forecasting with Stochastic Interpolants and Föllmer Processes. *ICML*, 2024.
- [3] Xinzhe Dai, Peichen Zhong, Bowen Deng, **Yifan Chen**, and Gerbrand Ceder. Inpainting crystal structure generations with score-based denoising. *ICML Workshop AI4Science*, 2024.
- [4] Zihui Wu, Yu Sun, **Yifan Chen**, Bingliang Zhang, Yisong Yue, Katherine L. Bouman. Principled Probabilistic Imaging using Diffusion Models as Plug-and-Play Priors. *NeurIPS*, 2024

GRANTS

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|--|-----------|
| • DataX Pilot Projects Grant Program, UCLA DataX | 2026 |
| • Research Allowance Program, UCLA Council on Research Faculty Grants | 2026 |
| • AMS-Simons Travel Grant, American Mathematical Society and Simons Foundation | 2025-2027 |

AWARDS

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| • The W.P. Carey and Co. Prize in Applied Mathematics, 2023, Caltech | 2023 |
| • Kortschak Scholars Program, Department of Computational and Mathematical Sciences, Caltech | 2018–2023 |
| • Tsinghua Xuetang Mathematics Program, Department of Mathematical Sciences, Tsinghua University | 2015–2018 |
| • Outstanding Undergraduate, Tsinghua University and Beijing | 2018 |
| • Baosteel Scholarship, Baosteel Corporation | 2017 |
| • Scholarship in Memory of the “12.9” Student Movement, Tsinghua University | 2016 |
| • Qualcomm Scholarship, Qualcomm Corporation | 2016 |
| • Scholarship in Memory of Mathematics Professor Ou Li, Tsinghua University | 2016 |
| • National Scholarship, Ministry of Education of China | 2015 |

TEACHING

Instructor at UCLA

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| • MATH 164: Optimization | Winter 2026 |
| • MATH 285J: Seminar in Applied Mathematics
(Topics in High-Dimensional Sampling and Generative Diffusions) | Spring 2026 |

Instructor at NYU Courant

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| • MATH-UA 120: Discrete Mathematics | Fall 2023 |
| • MATH-UA 120: Discrete Mathematics | Spring 2024 |
| • MATH-UA 252: Numerical Analysis | Fall 2024 |
| • MATH-UA 252: Numerical Analysis | Spring 2025 |

Teaching Assistant at Caltech

• ACM 109: Mathematical Modeling	Spring 2021
• ACM 118: Stochastic Processes and Regression	Winter 2020
• ACM 117: Probability and Stochastic Processes	Fall 2020
• ACM 109: Mathematical Modeling	Spring 2020

INDUSTRIAL EXPERIENCES

Citadel Securities	Miami, Florida
Quantitative Research Intern	6/2022-8/2022
– Project on Alpha Research: Predicting APAC Market Returns	
Microsoft	(virtual) Redmond, Washington
Part Time Researcher, working with Pengchuan Zhang	9/2021-2/2022
– Project: Stabilizing Large Scale Neural Network Training of Vision Transformers	

REFeree SERVICES

- **Journal:** Communications on Pure and Applied Mathematics, Annals of Applied Probability, Proceedings of the National Academy of Sciences (PNAS), Journal of Machine Learning Research, Mathematics of Computation, Journal of the Royal Statistical Society Series B, Journal of Computational Physics, Journal of Functional Analysis, SIAM on Uncertainty Quantification, SIAM on Imaging Sciences, SIAM on Optimization, SIAM on Mathematics of Data Science, SIAM on Multiscale Modeling and Simulation, SIAM on Numerical Analysis, SIAM on Control and Optimization, IMA Journal of Numerical Analysis, Analysis and Applications, Nature Machine Intelligence, Linear Algebra and Its Applications, European Journal of Applied Mathematics, Foundations of Data Science, Discrete and Continuous Dynamical Systems, Research in the Mathematical Sciences, Computational Methods in Applied Mathematics, Computers & Fluids, International Journal of Computer Mathematics
- **Conference:** NeurIPS (outstanding reviewer), International Conference on Geometric Science of Information (GSI)

CONFERENCES AND SEMINARS

- “Generative diffusion and sampling: algorithms and theory in high dimensions”, DataX+IDRE AI Symposium, UCLA, May 2026
- “Affine Invariant Samplers and Flows: Analysis and New Algorithms”, Monte Carlo online seminar, April 2026
- “Affine Invariant Samplers and Flows: Analysis and New Algorithms”, SIAM Conference on Uncertainty Quantification (UQ26), Minneapolis, Minnesota, March 2026
- “Variational Optimality of Föllmer Processes and related design questions in generative diffusions”, Generative modeling reading group seminar, Duke University (virtual), March 2026
- “Exploring High Dimensions in Dynamical Sampling: Flattening the Scaling Curve!”, Mathematics of Information, Data, and Signals Seminar, UCSD, San Diego, Feb 2026
- NeurIPS 2025, San Diego, Dec 2025
- “Exploring High Dimensions in Dynamical Sampling: Flattening the Scaling Curve!”, Applied Mathematics Seminar, University of California, Berkeley, Nov 2025
- “Exploring High Dimensions in Dynamical Sampling: Flattening the Scaling Curve!”, Applied Mathematics Colloquium, UCLA, Oct 2025
- “Exploring High Dimensions in Dynamical Sampling: Flattening the Scaling Curve!”, Scientific Computing and Numerics seminar, Cornell University (virtual), Oct 2025
- “Exploring High Dimensions in Dynamical Sampling: Flattening the Scaling Curve!”, Computational Math + X seminar, Caltech, Oct 2025

- “Exploring High Dimensions in Dynamical Sampling: Flattening the Scaling Curve!”, PDE & Applied Mathematics seminar, UC Riverside (virtual), Oct 2025
- “Exploring High Dimensions in Dynamical Sampling: Flattening the Scaling Curve!”, Computational and Applied Mathematics Colloquium, Penn State University, Sep 2025
- “Lipschitz Guided Dynamical Transport for generative flows of ill-conditioned distributions”, Workshop on Scientific Machine Learning: Theory, Algorithms, and Applications, Purdue University, Sep 2025
- “Convergence of Unadjusted Langevin in High Dimensions: Delocalization of Bias”, International Conference on Monte Carlo Methods and Applications, Illinois Institute of Technology, Chicago, July 2025
- “New Affine Invariant Ensemble Samplers and Their Dimensional Scaling”, Computational Bayesian Statistics Journal Club, Flatiron Institute, New York, May, 2025
- “Analysis and design of high dimensional sampling with applications in scientific computing ”, Purdue-PSU-Maryland joint mathematical data science seminar (virtual), April, 2025
- “Analysis and design of high dimensional sampling with applications in scientific computing ”, Applied Math Colloquium, UCLA, Los Angeles, Jan 2025
- “Convergence of Unadjusted Langevin in High Dimensions: Delocalization of Bias”, Probability Seminar, University of Michigan, Ann Arbor, Jan 2025
- “Analysis and design of high dimensional sampling with applications in scientific computing ”, Colloquium Seminar, University of Michigan, Ann Arbor, Jan 2025
- “Analysis and design of high dimensional sampling with applications in scientific computing ”, Applied Math Colloquium, Duke University, Durham, Jan 2025
- “Analysis and design of high dimensional sampling with applications in scientific computing ”, CMOR Special Lecture, Rice University, Houston, Dec 2024
- “Randomized and sparse Cholesky for learning with Gaussian processes”, SIAM Conference on Mathematics of Data Science (MDS24), Atlanta, Georgia, Oct 2024
- Recent Advances and Future Directions for Sampling, Yale University, New Haven, Oct 2024
- “On approximate, efficient sampling in high dimensional scientific computing”, Data Science Seminar, Department of Mathematics, University of Minnesota, Oct 2024
- “On approximate, efficient sampling in high dimensional scientific computing”, Rising Stars in Computing for Science and Engineering, Harvard University, Sep 2024
- “Convergence of Unadjusted Langevin in High Dimensions: Delocalization of Bias”, Computational Bayesian Statistics Journal Club, Flatiron Institute, New York, Sep, 2024
- “Fast, Multimodal, Derivative-Free Bayes Inference with Fisher-Rao Gradient Flows”, Applied Mathematics Seminar, Nanyang Technology University, July, 2024
- “Probabilistic Forecasting with Stochastic Interpolants and Föllmer Processes”, International Conference on Scientific Computation and Differential Equations, National University of Singapore, July, 2024
- Conference on Multiscale Modeling based on Physics and Data, IPAM, UCLA, April, 2024
- “Fast, Multimodal, Derivative-Free Bayes Inference with Fisher-Rao Gradient Flows”, Columbia Applied Math Colloquium, January, 2024
- “Fast, Multimodal, Derivative-Free Bayes Inference with Fisher-Rao Gradient Flows”, Workshop on Scientific Computing and Large Data, University of South Carolina, Dec, 2023
- “Fast, Multimodal, Derivative-Free Bayes Inference with Fisher-Rao Gradient Flows”, Numerical Analysis Seminar (virtual), Hong Kong University, Dec, 2023
- Measure Transport, Diffusion Processes and Sampling Workshop, Flatiron, New York, Dec, 2023
- “On Multiscale and Statistical Numerical Methods for PDEs and Inverse Problems”, Courant Postdoc day, Nov, 2023
- “Design of Gradient Flows for Sampling: Energy Functionals, Invariance, and Gaussian Approximation”, Yau Mathematical Science Center CAM seminar, Tsinghua University (virtual), Nov, 2023
- International Workshop on Recent Developments in Applied Mathematics and its Applications, Caltech, Nov, 2023

- “Design of Gradient Flows for Sampling: Energy Functionals, Invariance, and Gaussian Approximation”, Scientific machine learning seminar, Courant Institute, Oct, 2023
- “Solving and Learning Nonlinear PDEs with Gaussian Processes”, 17th U. S. National Congress on Computational Mechanics, Albuquerque, New Mexico, July 2023
- “Sparse Cholesky Factorization for Solving Nonlinear PDEs via Gaussian Processes”, Mathematical and Scientific Machine Learning, ICERM, Providence, June, 2023
- “Design of Gradient Flows for Sampling: Energy Functionals, Invariance, and Gaussian Approximation”, The AIMS Conference on Dynamical Systems, Differential Equations and Applications, Wilmington, North Carolina, May 2023
- “Gradient flows for sampling: affine invariance and numerical approximation”, Southern California Applied Mathematics Symposium, University of California, Irvine, April 2023
- “Design of Gradient Flows for Sampling: Energy Functionals, Invariance, and Gaussian Approximation”, Peking University applied math colloquium (virtual), Feb, 2023.
- “Developments of Multiscale and Probabilistic Methods for Solving PDEs and Inverse Problems”, Columbia applied math colloquium (virtual), January, 2023.
- The International Conference on New Trends in Computational and Data Sciences, Caltech, December 2022.
- “Solving and Learning Nonlinear PDEs with Gaussian Processes”, SIAM Mathematics of Data Science, San Diego, September, 2022.
- “Solving and Learning Nonlinear PDEs with Gaussian Processes”, Southern California Applied Mathematics Symposium, Harvey Mudd College, May, 2022
- “Solving and Learning Nonlinear PDEs with Gaussian Processes”, Rough Path Interest Group, The Alan Turing Institute (virtual), April, 2022
- “Solving and Learning Nonlinear PDEs with Gaussian Processes”, SIAM Uncertainty Quantification Minisymposium “New Developments in Gaussian Processes”, Atlanta, April 2022
- “Solving and Learning Nonlinear PDEs with Gaussian Processes”, CMX Student and Postdoc Seminar, Caltech, November, 2020
- “Consistency of Hierarchical Parameter Learning: Empirical Bayesian and Kernel Flow Approaches”, Second Symposium on Machine Learning and Dynamical Systems, Fields Institute (virtual), September, 2020
- “Consistency of Hierarchical Parameter Learning: Empirical Bayesian and Kernel Flow Approaches”, Bernoulli-IMS One World Symposium (virtual), August, 2020
- Oberwolfach Seminar: Beyond Numerical Homogenization, June, 2019
- Machine Learning for Multiscale Model Reduction Workshop, Harvard University, March, 2019
- Mathematical Model and Computation of Nonlinear Problems, Tsinghua Sanya International Mathematics Forum, January, 2018
- “The Quadratic Wasserstein Metric for Earthquake Location”, Youth Forum in the 15th Annual Meeting of CSIAM, Qingdao, China, October, 2017

ORGANIZATION OF TECHNICAL SESSIONS, WORKSHOPS, AND CONFERENCES

- Organizer committee of UCLA Math Colloquium
- Organizer committee of UCLA Data Theory Seminar
- Co-organizing the minisymposium “Analysis of Langevin and related algorithms” in International Conference on Monte Carlo Methods and Applications, Illinois Institute of Technology, Chicago, July 2025
- Co-organizing the minisymposium “Efficient computation and learning with randomized sampling and pruning” at the SIAM Conference on Mathematics of Data Science (MDS24), Atlanta, Georgia, Oct 2024
- Co-organizing the minisymposium “Recent Advances in Kernel Methods for Computing and Learning” in SIAM Mathematics of Data Science, San Diego, September, 2022.

COMPUTER SKILLS

Professional experiences in Python, Julia, MATLAB.

LANGUAGES

English (fluent), Chinese (native)